Topic 01: Inverse Laplace Transform

Unit V — Inverse Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

Contents

1 Opening Hook

Hook

All of Unit IV was about the forward journey: $f(t) \to F(s)$. The Laplace Transform is a machine that converts time-domain functions into s-domain representations. But what about the return trip? You solve an algebraic equation in the s-domain and get $Y(s)$. To get the actual solution $y(t)$, you need to invert — to press "undo" on the Laplace machine. The Inverse Laplace Transform is that undo button. This topic shows you exactly how it works, and gives you the standard inverse table that mirrors everything you learned in Unit IV.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this. Solving ordinary differential equations via Laplace transforms is a two-stage process: convert the ODE to an algebraic equation in s, solve for $Y(s)$, then *invert* to recover $y(t)$. Without the inverse step, you stop halfway. The forward transform gets you to $Y(s)$; the inverse transform brings you home to $y(t)$.

Mistake	**Consequence**
Not learning the inverse table	Stuck at $Y(s)$ — cannot get the actual ODE solution
Assuming the inverse is always unique	Missing the role of Lerch's theorem in guaranteeing uniqueness
Using $\mathcal{L}$ notation for both forward and inverse	Confusing direction — must use $\mathcal{L}^{-1}$

Key Point

The inverse Laplace transform is not optional — it is the step that converts your algebraic s-domain answer into the actual time-domain solution. Master the inverse table and you master the full Laplace technique.

3 You Already Know This (Intuition First)

Think of the Laplace transform as an encryption algorithm. In Unit IV, you fed a time-domain function $f(t)$ into the machine and received an encrypted message $F(s)$. The Inverse Laplace Transform is the *decryption key*.

Just as a table of ciphers lets you decode an encoded message letter by letter, the inverse transform table lets you decode $F(s)$ back to $f(t)$ entry by entry. You already know the entries from Unit IV — the inverse table is simply the same table read from right to left. There is no new machinery; there is only a new direction of travel.

For most engineering problems in this course, the workflow is:

1. Apply $\mathcal{L}$ to the ODE $\Rightarrow$ algebraic equation in s.

2. Solve for $Y(s)$ algebraically.
3. Apply $\mathcal{L}^{-1}$ using the inverse table (and partial fractions when $Y(s)$ is complicated) $\Rightarrow$ solution $y(t)$.

This topic covers Step 3.

4 Definition of Inverse Laplace Transform

Definition

If $\mathcal{L}\{f(t)\} = F(s)$, then the **Inverse Laplace Transform** of $F(s)$ is defined as

$$\mathcal{L}^{-1}\{F(s)\} = f(t),$$

meaning $f(t)$ is the function whose Laplace transform is $F(s)$.

Formal inversion integral (Bromwich / Mellin–Fourier):

$$f(t) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} e^{st}\,F(s)\,ds, \qquad t > 0,$$

where c is chosen so that all singularities of $F(s)$ lie to the left of the vertical line $\mathrm{Re}(s) = c$. In this course we do *not* evaluate this contour integral directly; instead, we use the inverse transform table together with algebraic techniques.

Linearity of the Inverse Transform:

$$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\,\mathcal{L}^{-1}\{F(s)\} + b\,\mathcal{L}^{-1}\{G(s)\}$$

for constants a, b.

Proof. Let $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$. Then, by linearity of $\mathcal{L}$,

$$\mathcal{L}\{af(t) + bg(t)\} = a\,\mathcal{L}\{f(t)\} + b\,\mathcal{L}\{g(t)\} = aF(s) + bG(s).$$

Applying $\mathcal{L}^{-1}$ to both sides gives $\mathcal{L}^{-1}\{aF(s) + bG(s)\} = af(t) + bg(t)$. $\square$

5 Lerch's Theorem — Uniqueness

Theorem (Lerch)

If $f_1(t)$ and $f_2(t)$ are both piecewise continuous on $[0, \infty)$ and

$$\mathcal{L}\{f_1\} = \mathcal{L}\{f_2\} = F(s),$$

then $f_1(t) = f_2(t)$ at all points of continuity of f_1 and f_2.

Interpretation: Among all piecewise continuous functions on $[0, \infty)$, the inverse Laplace transform is *unique*. The notation $\mathcal{L}^{-1}\{F(s)\} = f(t)$ is therefore well-defined. Two piecewise continuous functions that share the same Laplace transform must be identical at every point where they are continuous. This theorem justifies looking up the inverse in a table rather than worrying about whether a different function might also qualify.

Why Lerch Matters

Lerch's theorem is the guarantee that underpins the entire inverse-transform strategy. Without uniqueness, the table lookup would be ambiguous. With it, you can be confident: *the* $f(t)$ you find from the table is the correct one.

6 Standard Inverse Transform Table

$F(s)$	$\mathcal{L}^{-1}\{F(s)\} = f(t)$	Validity condition
$\dfrac{1}{s}$	1	$s > 0$
$\dfrac{1}{s^n}$	$\dfrac{t^{n-1}}{(n-1)!}, \quad n \geq 1$	$s > 0$
$\dfrac{1}{s-a}$	e^{at}	$s > a$
$\dfrac{s}{s^2+a^2}$	$\cos(at)$	$s > 0$
$\dfrac{a}{s^2+a^2}$	$\sin(at)$	$s > 0$
$\dfrac{s}{s^2-a^2}$	$\cosh(at)$	$s > \lvert a \rvert$
$\dfrac{a}{s^2-a^2}$	$\sinh(at)$	$s > \lvert a \rvert$
$\dfrac{e^{-as}}{s}$	$u(t-a)$	$s > 0,\ a \geq 0$
e^{-as}	$\delta(t-a)$	$a \geq 0$
$F(s-a)$	$e^{at}f(t)$	(s-shift, inverse form)

Here $u(t-a)$ denotes the unit step function and $\delta(t-a)$ denotes the Dirac delta.

7 Inverse of the First Shifting Theorem

Theorem — Inverse First Shifting

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, then

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t).$$

That is, a shift of a in the s-variable corresponds to multiplication by e^{at} in the time domain.

Worked Application: Find $\mathcal{L}^{-1}\left\{\dfrac{1}{(s-3)^2}\right\}$.

Step 1 — Identify the base form. Write $F(s) = \dfrac{1}{s^2}$. From the standard table,

$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$, so $f(t) = t$.

Step 2 — Recognise the shift. The given expression is $F(s-3)$, i.e. the base form evaluated at $s-3$ instead of s, with $a=3$.

Step 3 — Apply the theorem.

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} = e^{3t}\,\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = e^{3t}\cdot t = t\,e^{3t}.$$

$F(s-a)$ form	Matched $F(u)$ (with $u=s-a$)	Result $e^{at}f(t)$
$\frac{1}{(s-3)^2}$	$\frac{1}{u^2} \to t$	$t\,e^{3t}$
$\frac{1}{(s+2)^3}$	$\frac{1}{u^3} \to \frac{t^2}{2}$	$\frac{t^2}{2}\,e^{-2t}$
$\frac{s-1}{(s-1)^2+9}$	$\frac{u}{u^2+9} \to \cos(3t)$	$e^t\cos(3t)$
$\frac{4}{(s+5)^2+16}$	$\frac{4}{u^2+16} \to \sin(4t)$	$e^{-5t}\sin(4t)$
$\frac{1}{s-a}$	$\frac{1}{u} \to 1$	e^{at}

What Did We Learn?

The inverse first shifting theorem is a pattern-matching tool: replace s with $s-a$ everywhere in a known $F(s)$, and the time-domain result is multiplied by e^{at}. Always identify the shift a and the base form $F(s)$ before applying the theorem.

8 Completing the Square

Technique — Completing the Square

When the denominator contains a quadratic s^2+bs+c that does not factor neatly, complete the square:

$$s^2+bs+c = \left(s+\frac{b}{2}\right)^2 + \left(c-\frac{b^2}{4}\right).$$

Once in this form, a shift $a=-b/2$ lets you match the standard sinusoidal or hyperbolic entries.

Full Worked Example: Find $\mathcal{L}^{-1}\left\{\frac{1}{s^2+4s+5}\right\}$.

Step 1 — Complete the square in the denominator.

$$s^2+4s+5 = (s+2)^2+1.$$

Step 2 — Rewrite the expression.

$$\frac{1}{s^2+4s+5} = \frac{1}{(s+2)^2+1^2}.$$

Step 3 — Match to standard form. Recognise $\dfrac{a}{(s-(-2))^2+a^2}$ with $a=1$ (numerator is $1=a$) and the shift -2 (i.e. replace s by $s-(-2)=s+2$).

Step 4 — Apply inverse shifting theorem.

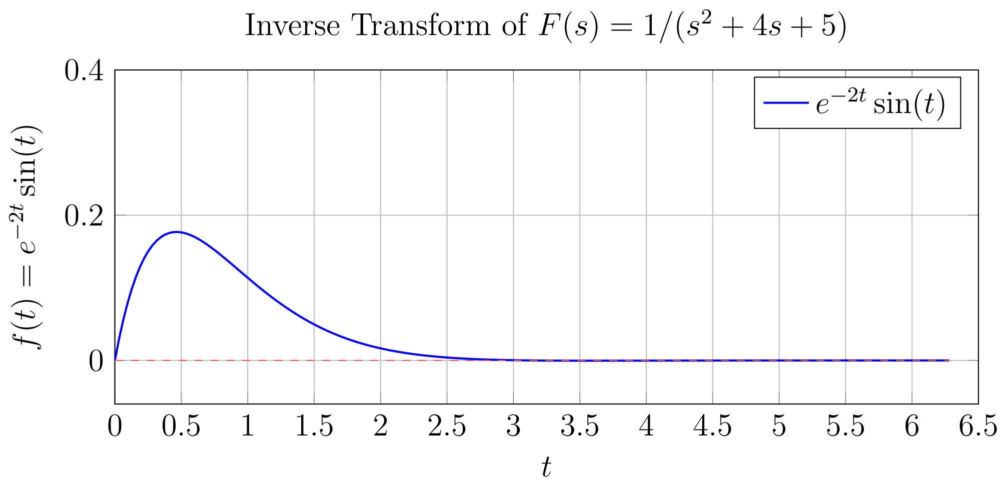
$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+1}\right\} = e^{-2t}\,\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = e^{-2t}\sin(t).$$

What Did We Learn?

Completing the square converts a messy quadratic denominator into a shifted standard form. Once the denominator is $(s-\alpha)^2+\omega^2$, you immediately match $\sin(\omega t)$ or $\cos(\omega t)$ and multiply by $e^{\alpha t}$.

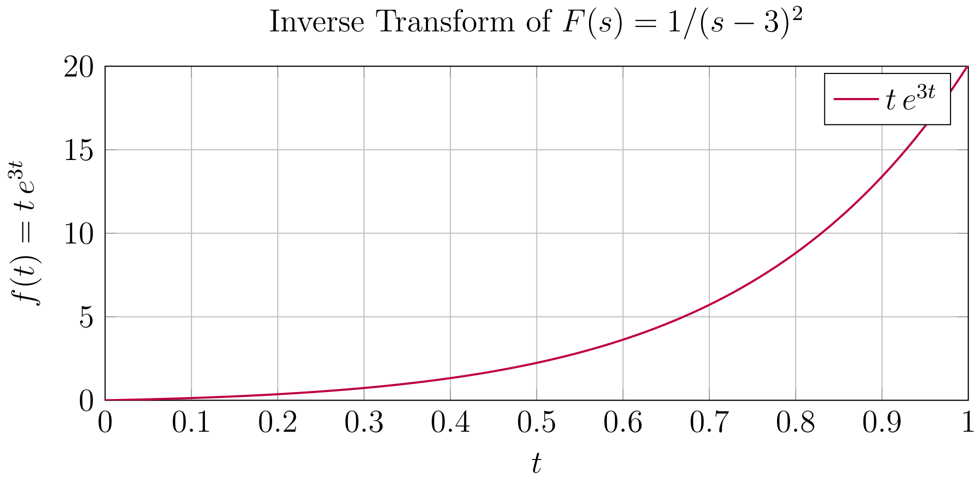
9 Visualizing the Inverse Transform

Plot 1: $f(t) = e^{-2t}\sin(t)$ — Damped Sinusoidal Response



This is the time-domain function corresponding to $F(s) = 1/(s^2+4s+5)$ — a damped sinusoidal response, typical of underdamped systems. The amplitude decays exponentially as e^{-2t} while oscillating at unit frequency.

Plot 2: $f(t) = t\,e^{3t}$ — Exponential Growth



This is the time-domain function corresponding to $F(s) = 1/(s-3)^2$, showing rapid exponential growth. Such responses appear in unstable systems where the s-domain pole lies in the right half-plane.

10 Worked Examples

Example 1: Linearity and the Standard Table

Find $\mathcal{L}^{-1}\left\{\frac{3}{s} + \frac{2}{s^3} - \frac{5}{s+4}\right\}$.

Step 1 — Apply linearity.

$$\mathcal{L}^{-1}\left\{\frac{3}{s} + \frac{2}{s^3} - \frac{5}{s+4}\right\} = 3\,\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 2\,\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} - 5\,\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\}.$$

Step 2 — Look up each term in the standard table.

- $\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1.$
- $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{t^2}{2!} = \frac{t^2}{2}$ (using $1/s^n \to t^{n-1}/(n-1)!$ with $n = 3$).
- $\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} = e^{-4t}$ (using $1/(s-a) \to e^{at}$ with $a = -4$).

Step 3 — Combine.

$$\mathcal{L}^{-1}\left\{\frac{3}{s} + \frac{2}{s^3} - \frac{5}{s+4}\right\} = 3 \cdot 1 + 2 \cdot \frac{t^2}{2} - 5\,e^{-4t} = 3 + t^2 - 5\,e^{-4t}.$$

What Did We Learn?

Linearity lets you invert term by term. For $1/s^n$, the formula $t^{n-1}/(n-1)!$ works for all $n \geq 1$. For $1/(s+4)$, read the shift as $a = -4$ and apply e^{at}.

Example 2: Matching Shifted Standard Forms

Find $\mathcal{L}^{-1}\left\{\dfrac{2s+1}{(s-1)^2+4}\right\}$.

Step 1 — Split the numerator to match standard entries. Write $2s+1$ in terms of $(s-1)$:

$$2s+1 = 2(s-1)+2+1 = 2(s-1)+3.$$

So

$$\frac{2s+1}{(s-1)^2+4} = \frac{2(s-1)}{(s-1)^2+4} + \frac{3}{(s-1)^2+4}.$$

Step 2 — Identify base forms and shifts.

- $\dfrac{2(s-1)}{(s-1)^2+4}$: matches $\dfrac{2u}{u^2+4}$ with $u = s-1$, i.e. $2 \cdot \dfrac{u}{u^2+2^2}$. Inverse: $\mathcal{L}^{-1}\left\{\dfrac{u}{u^2+4}\right\} = \cos(2t)$. With shift $a = 1$: gives $2\,e^t\cos(2t)$.
- $\dfrac{3}{(s-1)^2+4}$: matches $\dfrac{3}{u^2+2^2}$. Write as $\dfrac{3}{2} \cdot \dfrac{2}{u^2+2^2}$. Inverse: $\dfrac{3}{2}\mathcal{L}^{-1}\left\{\dfrac{2}{u^2+4}\right\} = \dfrac{3}{2}\sin(2t)$. With shift $a = 1$: gives $\dfrac{3}{2}e^t\sin(2t)$.

Step 3 — Combine.

$$\mathcal{L}^{-1}\left\{\frac{2s+1}{(s-1)^2+4}\right\} = e^t\left(2\cos(2t) + \frac{3}{2}\sin(2t)\right).$$

What Did We Learn?

When a numerator contains s and the denominator is $(s-a)^2+\omega^2$, always rewrite the numerator as a multiple of $(s-a)$ plus a remainder. This splits the fraction cleanly into a cosine-type and a sine-type term.

Example 3: Completing the Square in the Denominator

Find $\mathcal{L}^{-1}\left\{\dfrac{s+3}{s^2+6s+13}\right\}$.

Step 1 — Complete the square in the denominator.

$$s^2+6s+13 = (s+3)^2+4.$$

Step 2 — Rewrite the expression.

$$\frac{s+3}{s^2+6s+13} = \frac{s+3}{(s+3)^2+4}.$$

Step 3 — Match to base form with shift. Let $u = s + 3$ (i.e. $a = -3$). Then

$$\frac{s+3}{(s+3)^2+4} = \frac{u}{u^2+2^2},$$

which matches $\dfrac{u}{u^2+\omega^2}$ with $\omega = 2$.

Step 4 — Apply the inverse transform.

$$\mathcal{L}^{-1}\left\{\frac{u}{u^2+4}\right\} = \cos(2t).$$

With shift $a = -3$:

$$\mathcal{L}^{-1}\left\{\frac{s+3}{s^2+6s+13}\right\} = e^{-3t}\cos(2t).$$

What Did We Learn?

When the numerator exactly matches the form $(s-\alpha)$ after completing the square, the result is a pure cosine shifted by $e^{\alpha t}$. No fractional splitting is needed — the numerator is already in the right form.

11 Excel Example (Numerical Verification)

We verify the inverse transform $f(t) = e^{-2t}\sin(t)$ numerically by confirming that the Laplace integral $\int_0^\infty e^{-st} f(t)\,dt$ evaluated at $s = 3$ gives $F(3) = 1/(9+12+5) = 1/26 \approx 0.0385$.

Row	**A:** t	**B:** $f(t) = e^{-2t}\sin(t)$	**C: integrand** $e^{-3t}f(t)$	**D: Cumulative trap. sum**
1	0.0	0.000000	0.000000	0.000000
2	0.1	0.081737	0.060552	0.003028
3	0.2	0.133172	0.073086	0.009710
4	0.3	0.162185	0.065939	0.016661
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
80	7.9	≈ 0	≈ 0	≈ 0.0385

Column formulas (starting at row 2, t in column A):

- Column A (time steps, 0 to 8 in steps of 0.1): Row 1 = 0; subsequent rows = `A`$(n-1)+0.1$.
- Column B: `=EXP(-2*A2)*SIN(A2)`
- Column C (integrand for $s = 3$): `=EXP(-3*A2)*EXP(-2*A2)*SIN(A2)` which simplifies to `=EXP(-5*A2)*SIN(A2)`
- Column D (cumulative trapezoidal sum): Row 1 = 0; rows ≥ 2: `=D1+(C1+C2)/2*0.1`

In row 80 (at $t = 7.9$), the cumulative sum converges to approximately 0.0385, matching the exact value $F(3) = 1/26 \approx 0.03846$ to four decimal places.

What Did We Learn?

The trapezoidal rule confirms numerically that $\int_0^\infty e^{-3t}e^{-2t}\sin(t)\,dt \approx 0.0385 = 1/26$. This validates $\mathcal{L}^{-1}\{1/(s^2+4s+5)\} = e^{-2t}\sin(t)$ beyond the symbolic table lookup. The integrand decays rapidly because the combined exponential e^{-5t} forces the integrand to zero well before $t = 8$.

12 Python Example

Listing 1: Symbolic Inverse Laplace Transforms with SymPy

```python
import sympy as sp

s, t = sp.symbols('s t')

# --- Example 1 ---
F1 = sp.Rational(3,1)/s + sp.Rational(2,1)/s**3 - sp.Rational(5,1)/(s+4)
f1 = sp.inverse_laplace_transform(F1, s, t)
print("Example 1:", sp.simplify(f1))
# Expected: 3 + t**2 - 5*exp(-4*t)  (for t >= 0)

# Verify by forward transform
F1_check = sp.laplace_transform(f1, t, s, noconds=True)
print("Verify Example 1 F(s):", sp.simplify(F1_check - F1) == 0)
# Expected: True

# --- Example 2 ---
F2 = (2*s + 1) / ((s-1)**2 + 4)
f2 = sp.inverse_laplace_transform(F2, s, t)
print("Example 2:", sp.simplify(f2))
# Expected: exp(t)*(2*cos(2*t) + (3/2)*sin(2*t))  (for t >= 0)

# Verify
F2_check = sp.laplace_transform(f2, t, s, noconds=True)
print("Verify Example 2 F(s):", sp.simplify(F2_check - F2) == 0)
# Expected: True

# --- Example 3 ---
F3 = (s + 3) / (s**2 + 6*s + 13)
f3 = sp.inverse_laplace_transform(F3, s, t)
print("Example 3:", sp.simplify(f3))
# Expected: exp(-3*t)*cos(2*t)  (for t >= 0)

# Verify
F3_check = sp.laplace_transform(f3, t, s, noconds=True)
print("Verify Example 3 F(s):", sp.simplify(F3_check - F3) == 0)
# Expected: True
```

What Did We Learn?

SymPy's `inverse_laplace_transform` handles all three techniques automatically. Re-applying the forward transform and checking that the difference is zero gives a clean symbolic verification. This workflow — compute inverse, verify forward — is a reliable sanity-check for any inverse transform computation.

13 Viva-Style Oral Questions

1. **Q: Define the inverse Laplace transform.**
   A: If $\mathcal{L}\{f(t)\} = F(s)$, then the inverse Laplace transform is $\mathcal{L}^{-1}\{F(s)\} = f(t)$. It returns a function of t whose Laplace transform is the given $F(s)$.

2. **Q: State Lerch's theorem and explain its significance.**
   A: Lerch's theorem states that if two piecewise continuous functions $f_1(t)$ and $f_2(t)$ on $[0, \infty)$ have the same Laplace transform $F(s)$, then $f_1(t) = f_2(t)$ at every point of continuity. Its significance: the inverse transform is unique among piecewise continuous functions, so the table-lookup approach is unambiguous.

3. **Q: What is the Bromwich integral and why is it not used directly in this course?**
   A: The Bromwich (Mellin–Fourier) integral is $f(t) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} e^{st}F(s)\,ds$. It is the rigorous inversion formula but requires contour integration in the complex plane (residues, etc.), which is beyond the scope of this B.Tech course. Instead, we use the inverse transform table and algebraic techniques.

4. **Q: State the linearity property of the inverse Laplace transform.**
   A: For constants a and b: $\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\,\mathcal{L}^{-1}\{F(s)\} + b\,\mathcal{L}^{-1}\{G(s)\}$. This follows directly from the linearity of the forward transform.

5. **Q: How do you use the inverse first shifting theorem?**
   A: If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, then $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$. In practice: identify the shift a, strip it to obtain the base form $F(s)$, look up $f(t) = \mathcal{L}^{-1}\{F(s)\}$ from the table, then multiply by e^{at}.

6. **Q: When and why do we complete the square?**
   A: We complete the square when the denominator is an irreducible quadratic $s^2 + bs + c$ that cannot be factored over the reals. The technique converts it to $(s-\alpha)^2 + \omega^2$, enabling a direct match to the sinusoidal entries in the inverse table combined with the shift theorem.

7. **Q: Give the inverse transforms of $1/s$, $1/s^2$, and $1/(s^2 + a^2)$.**
   A: $\mathcal{L}^{-1}\{1/s\} = 1$; $\quad \mathcal{L}^{-1}\{1/s^2\} = t$; $\quad \mathcal{L}^{-1}\left\{\dfrac{1}{s^2 + a^2}\right\} = \dfrac{1}{a}\sin(at)$ (since $\mathcal{L}^{-1}\{a/(s^2 + a^2)\} = \sin(at)$).

8. **Q: What notation distinguishes the forward and inverse Laplace transforms?**
   A: $\mathcal{L}$ denotes the forward (direct) Laplace transform: $f(t) \mapsto F(s)$. $\mathcal{L}^{-1}$ denotes the inverse Laplace transform: $F(s) \mapsto f(t)$. These must never be confused; using the wrong symbol reverses the direction of the operation.

14 Descriptive Questions

1. **Define the inverse Laplace transform and state Lerch's theorem.**
Write a complete answer covering: (i) the definition $\mathcal{L}^{-1}\{F(s)\} = f(t)$ means $f(t)$ is the function whose Laplace transform is $F(s)$; (ii) the Bromwich integral as the formal inversion formula; (iii) Lerch's theorem: uniqueness among piecewise continuous functions on $[0, \infty)$; (iv) why uniqueness justifies the table-based approach.

2. **Prove that the inverse Laplace transform is linear.**
Let $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$. By linearity of $\mathcal{L}$: $\mathcal{L}\{af(t)+bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\} = aF(s) + bG(s)$. Applying $\mathcal{L}^{-1}$: $\mathcal{L}^{-1}\{aF(s) + bG(s)\} = af(t) + bg(t)$. This proof is complete.

3. **Find $\mathcal{L}^{-1}\left\{\dfrac{2s+3}{s^2+4s+7}\right\}$ step by step.**
(i) Complete the square: $s^2+4s+7 = (s+2)^2+3$. (ii) Split the numerator: $2s+3 = 2(s+2)-4+3 = 2(s+2)-1$. (iii) Write: $\dfrac{2(s+2)-1}{(s+2)^2+3} = \dfrac{2(s+2)}{(s+2)^2+3} - \dfrac{1}{(s+2)^2+3}$.
(iv) Match and apply shift ($a = -2$, $\omega = \sqrt{3}$): $2e^{-2t}\cos(\sqrt{3}\,t) - \dfrac{1}{\sqrt{3}}e^{-2t}\sin(\sqrt{3}\,t)$.

4. **State and apply the inverse first shifting theorem.**
State: if $\mathcal{L}^{-1}\{F(s)\} = f(t)$ then $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$. Application: find $\mathcal{L}^{-1}\{1/(s-2)^4\}$. Base form $F(s) = 1/s^4$: $\mathcal{L}^{-1} = t^3/6$. Shift $a = 2$: $\mathcal{L}^{-1}\{1/(s-2)^4\} = \dfrac{t^3}{6}e^{2t}$.

5. **Explain the relationship between the forward and inverse tables.**
The forward table lists $f(t) \to F(s)$. Reading the same table in reverse gives the inverse table $F(s) \to f(t)$. Both tables carry identical information; the direction of lookup is all that changes. Lerch's theorem guarantees that the reverse lookup is unambiguous for piecewise continuous functions. In practice, every entry learned in Unit IV is automatically an entry in the Unit V inverse table.

15 Practice Problems

1. Find $\mathcal{L}^{-1}\left\{\dfrac{4}{s^3}\right\}$.
Hint: Use $1/s^n \to t^{n-1}/(n-1)!$ with $n = 3$.
Answer: $2t^2$.

2. Find $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2-9}\right\}$.
Hint: Write as $\dfrac{1}{(s-3)(s+3)}$ and use $\sinh(at)/a \leftrightarrow 1/(s^2-a^2)$ or partial fractions.
Answer: $\dfrac{1}{3}\sinh(3t)$.

3. Find $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2+4s+8}\right\}$.
Hint: Complete the square: $s^2+4s+8 = (s+2)^2+4$. Split $s = (s+2)-2$.
Answer: $e^{-2t}(\cos(2t) - \sin(2t))$.

4. Find $\mathcal{L}^{-1}\left\{\dfrac{3}{(s-2)^4}\right\}$.
   Hint: Use $1/s^4 \to t^3/6$, then shift $a = 2$.
   Answer: $\dfrac{t^3}{2}\,e^{2t}$.

5. Find $\mathcal{L}^{-1}\left\{\dfrac{e^{-2s}}{s}\right\}$.
   Hint: $e^{-as}/s \to u(t-a)$ from the standard table.
   Answer: $u(t-2)$ (unit step shifted to $t = 2$).

6. Find $\mathcal{L}^{-1}\left\{\dfrac{s+1}{s^2+2s+5}\right\}$.
   Hint: Complete the square: $s^2 + 2s + 5 = (s+1)^2 + 4$. Numerator $s + 1$ matches $(s - (-1))$ exactly.
   Answer: $e^{-t}\cos(2t)$.

16 MCQs

1. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}^{-1}\{F(s)\}$ equals:
   (a) $F(t)$ (b) $f(-t)$ **(c)** $f(t)$ (d) $F(-s)$
   Explanation: By definition, $\mathcal{L}^{-1}$ inverts the Laplace transform, returning the original time-domain function $f(t)$.

2. Lerch's theorem guarantees that the inverse Laplace transform is:
   (a) Always zero for $t < 0$ (b) Differentiable everywhere **(c) Unique among piecewise continuous functions** (d) Defined only for rational $F(s)$
   Explanation: Lerch's theorem ensures uniqueness among piecewise continuous functions on $[0, \infty)$, making the table-lookup unambiguous.

3. $\mathcal{L}^{-1}\left\{\dfrac{1}{s}\right\}$ equals:
   (a) t **(b)** 1 (c) e^t (d) $\delta(t)$
   Explanation: From the standard table, $\mathcal{L}\{1\} = 1/s$, so $\mathcal{L}^{-1}\{1/s\} = 1$.

4. The inverse first shifting theorem states $\mathcal{L}^{-1}\{F(s-a)\}$ equals:
   (a) $f(t-a)$ (b) $f(t)u(t-a)$ **(c)** $e^{at}f(t)$ (d) $e^{-at}f(t)$
   Explanation: Shifting the argument from s to $s - a$ in the s-domain corresponds to multiplication by e^{at} in the time domain.

5. Completing the square in the denominator is used to:
   (a) Simplify the numerator polynomial (b) Convert the transform to partial fractions **(c) Match the quadratic denominator to the standard sinusoidal forms** (d) Eliminate complex roots
   Explanation: Completing the square converts $s^2 + bs + c$ to $(s-\alpha)^2 + \omega^2$, enabling a direct match to the $\sin(\omega t)$ or $\cos(\omega t)$ entries in the inverse table via the shift theorem.

17 Common Mistakes

Common Mistakes in Inverse Laplace Transforms

Mistake	What Students Do	Correct Approach
Confusing $\mathcal{L}$ with $\mathcal{L}^{-1}$	Write $\mathcal{L}$ when inverting, or apply the forward table in the wrong direction	Always write $\mathcal{L}^{-1}\{F(s)\}$ when going from s-domain to t-domain
Skipping completing the square	Try to invert $1/(s^2+4s+5)$ directly by guessing or wrong factoring	First complete the square: $s^2+4s+5=(s+2)^2+1$; then match
Matching $1/(s^2+a^2)$ to $\sin(at)$ incorrectly	Write $\mathcal{L}^{-1}\{1/(s^2+a^2)\}=\sin(at)$ without checking the numerator	The correct entry is $\mathcal{L}^{-1}\{a/(s^2+a^2)\}=\sin(at)$; the numerator must be a, not 1
Applying the shift to only part of the expression	When finding $\mathcal{L}^{-1}\{F(s-a)\}$, substitute $s-a$ into the formula but forget to replace every occurrence of s	Replace s by $s-a$ everywhere in $F(s)$ before looking up the base form
Assuming $\mathcal{L}^{-1}\{F(s)G(s)\}=f(t)g(t)$	Multiply the two inverse transforms directly	The correct result is the convolution $(f*g)(t)=\int_0^t f(\tau)g(t-\tau)\,d\tau$ (covered in Topic 04)

18 Quick Recap

Quick Recap — Topic 01: Inverse Laplace Transform

- $\mathcal{L}^{-1}\{F(s)\}=f(t)$ is the function whose Laplace transform is $F(s)$ — it is the "undo" operation.
- The formal inversion is the Bromwich integral; in practice, use the inverse table and algebraic techniques.
- **Lerch's theorem**: the inverse is unique among piecewise continuous functions on $[0,\infty)$.
- **Linearity**: $\mathcal{L}^{-1}\{aF(s)+bG(s)\}=af(t)+bg(t)$ — invert term by term.
- **Standard inverse table**: 10 core entries covering powers, exponentials, sinusoids, hyperbolic functions, unit step, and delta.
- **Inverse first shifting theorem**: $\mathcal{L}^{-1}\{F(s-a)\}=e^{at}f(t)$ — a shift in s by a multiplies $f(t)$ by e^{at}.
- **Completing the square**: converts s^2+bs+c to $(s-\alpha)^2+\omega^2$, enabling direct match to sinusoidal table entries.

- Up next (Topic 02): the full inverse table for elementary functions; Topics 03–04: partial fractions and convolution for complex $F(s)$.